Doiraning yuzi

1-test (oson) · 20 ta savol

Ism-familiya: _____ Sinf: _____ Sana: _____

Har savolda bitta to'g'ri javobni belgilang.

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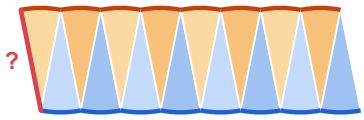
1. **Doiraning yuzi qaysi formula bilan topiladi?**

A) 2πr **B)** πd **C)** 2πr² **D)** πr²

2. **r² yozuvi nimani bildiradi?**

A) 2 · r **B)** r : 2 **C)** r · r **D)** r + r

3. **Doira teng bo'laklarga kesilib, rasmdagidek qatorga terildi. «?» bilan belgilangan qizil kesma nimaga teng?**



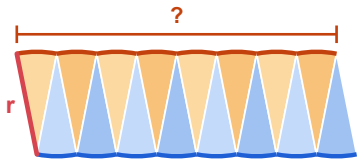
A) radiusga

B) aylana uzunligiga

C) aylana uzunligining yarmiga

D) diametrga

4. **Doira teng bo'laklarga kesilib, rasmdagidek qatorga terildi. Shaklning «?» bilan belgilangan eni nimaga teng?**



A) aylana uzunligiga

B) diametrga

C) aylana uzunligining yarmiga

D) radiusga

5. **Radiusi 8 cm bo'lgan doiraning yuzini π orqali ifodalang.**

A) 16π cm² **B)** 32π cm² **C)** 8π cm² **D)** 64π cm²

6. **Diametri 14 dm bo'lgan doiraning yuzini π orqali ifodalang.**

A) 14π dm² **B)** 196π dm² **C)** 49π dm² **D)** 28π dm²

7. **Radiusi 10 cm bo'lgan doiraning yuzini toping (π = 3,14).**

A) 31,4 cm² **B)** 62,8 cm² **C)** 628 cm² **D)** 314 cm²

8. **Radiusi 3 dm bo'lgan doiraning yuzini toping (π = 3,14).**

A) 28,26 dm² **B)** 9,42 dm²

C) 113,04 dm² **D)** 18,84 dm²

9. **Radiusi 4,2 m bo'lgan doiraning yuzini toping (π = $\frac{22}{7}$).**

A) 55,44 m² **B)** 26,4 m² **C)** 13,2 m² **D)** 221,76 m²

10. **Diametri 12 m bo'lgan doiraning yuzini toping (π = 3,14).**

A) 113,04 m² **B)** 452,16 m² **C)** 18,84 m² **D)** 37,68 m²

11. Halqaning yuzi qanday topiladi?

A) π · (katta radius − kichik radius)

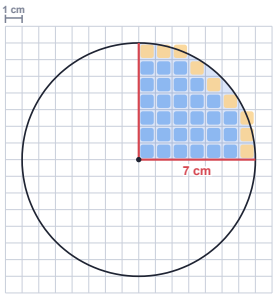
B) katta radius − kichik radius

C) katta doira yuzi + kichik doira yuzi

D) katta doira yuzi − kichik doira yuzi

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12. Rasmda radiusi 7 cm bo'lgan doiraning choragi kataklarga bo'lingan (1 katak — 1 cm²). Ko'k — to'liq kataklar (30 ta), sariq — yarmidan ko'pi chorak ichida bo'lgan kataklar (9 ta). Butun doiraning yuzi taxminan qancha?



A) 196 cm²

B) 39 cm²

C) 120 cm²

D) 156 cm²

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13. Radiusi r bo'lgan doira tomoni 2r bo'lgan kvadrat ichiga joylashgan. Kvadratning yuzi qancha?

A) 2r²

B) 3r²

C) 4r²

D) r²

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14. Radiusi r bo'lgan doiraning yuzi haqida qaysi gap to'g'ri?

A) 4r² dan katta

B) aynan 4r² ga teng

C) 2r² dan katta, 4r² dan kichik

D) 2r² dan kichik

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15. Ali diametri 18 cm bo'lgan doiraning yuzini 3,14 · 18 · 18 = 1017,36 cm² deb topdi. U qanday xatoga yo'l qo'ydi?

A) xato yo'q

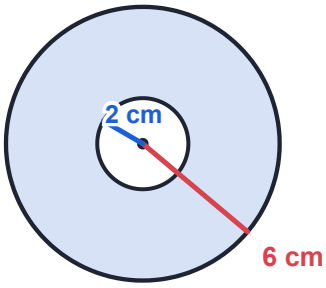
B) radius o'rniga diametrni oldi

C) π ni noto'g'ri oldi

D) r² o'rniga 2r ni oldi

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16. Bo'yalgan halqaning yuzini toping (π = 3,14).



A) 125,6 cm²

B) 100,48 cm²

C) 50,24 cm²

D) 113,04 cm²

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17. Radiusi 1,5 m bo'lgan doiraning yuzini toping (π = 3,14).

A) 4,71 m²

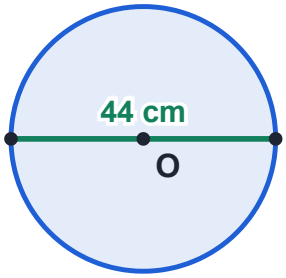
B) 9,42 m²

C) 7,065 m²

D) 28,26 m²

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18. O — doira markazi. Doiraning yuzini toping (π = 3,14).



A) 69,08 cm²

B) 138,16 cm²

C) 6079,04 cm²

D) 1519,76 cm²

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19. **Diametri 5,6 m bo'lgan doiraning yuzini toping ($\pi = \frac{22}{7}$).**

A) 98,56 m² **B)** 8,8 m² **C)** 24,64 m² **D)** 17,6 m²

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20. **Radiusi 4 cm bo'lgan doira bo'laklaridan deyarli to'g'ri to'rtburchak tuzildi. Uning eni (aylana uzunligining yarmi) qancha? (π = 3,14)**

A) 12,56 cm **B)** 50,24 cm **C)** 25,12 cm **D)** 4 cm

Javoblar

Doiraning yuzi · 1-test (oson)

№	Javob	Qiymati
1	**D**	πr²
2	**C**	r · r
3	**A**	radiusga
4	**C**	aylana uzunligining yarmiga
5	**D**	64π cm²
6	**C**	49π dm²
7	**D**	314 cm²
8	**A**	28,26 dm²
9	**A**	55,44 m²
10	**A**	113,04 m²

№	Javob	Qiymati
11	**D**	katta doira yuzi − kichik doira yuzi
12	**D**	156 cm²
13	**C**	4r²
14	**C**	2r² dan katta, 4r² dan kichik
15	**B**	radius o'rniga diametrni oldi
16	**B**	100,48 cm²
17	**C**	7,065 m²
18	**D**	1519,76 cm²
19	**C**	24,64 m²
20	**A**	12,56 cm

Qisqa yechimlar

1. Doira yuzi = π · radius · radius.
Qisqacha: **πr²**. (2πr va πd — aylana uzunligi.)

2. Kvadrat — sonni o'ziga ko'paytirish.
r² = **r · r**. Masalan, 5² = 5 · 5 = 25.

3. Har bir bo'lakning yon tomoni doira markazidan aylanagacha boradi.
Demak, u **radiusga** teng.

4. Aylananing yarmi tepaga, ikkinchi yarmi pastga tushdi.
Eni — **aylana uzunligining yarmi**.

5. Yuzi = π · 8 · 8 = **64π cm²**.
(16π — aylana uzunligi: 2 · π · 8.)

6. Avval radius: 14 : 2 = 7 dm.
Yuzi = π · 7 · 7 = **49π dm²**.
(196π — radius o'rniga diametr olingan.)

7. Yuzi = 3,14 · 10 · 10 = 3,14 · 100 = **314 cm²**.

8. Yuzi = 3,14 · 3 · 3 = 3,14 · 9 = **28,26 dm²**.
(18,84 — r² o'rniga 2r olingan.)

9. $\frac{22}{7}$ · 4,2 · 4,2. Avval 4,2 : 7 = 0,6.
22 · 0,6 · 4,2 = **55,44 m²**.

10. Radius: 12 : 2 = 6 m.
Yuzi = 3,14 · 6 · 6 = **113,04 m²**.
(452,16 — diametr radius deb olingan.)

11. Halqa — katta doiradan kichik doira olib tashlangandan keyin qolgan qism.
Yuzi = **katta doira yuzi − kichik doira yuzi**.

12. Ko'k kataklar 30 ta, sariqlari 9 ta: chorak ≈ 30 + 9 = 39 cm².
Doira 4 ta chorakdan iborat: 4 · 39 = **156 cm²**.

13. Kvadrat yuzi = 2r · 2r = **$4r^2$**.
Doira undan kichik.

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14. Doira ichidagi kvadrat yuzi $2r^2$, doirani o'rab turgan kvadrat yuzi $4r^2$.
Doira yuzi ular orasida: **$2r^2$ dan katta, $4r^2$ dan kichik** (≈ $3{,}14r^2$).

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15. Formulada radius turadi: r = 18 : 2 = 9 cm.
To'g'risi: 3,14 · 9 · 9 = 254,34 cm^2. Ali **radius o'rniga diametrni oldi**.

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16. Katta doira: 3,14 · 6 · 6 = 113,04 cm^2.
Kichik doira: 3,14 · 2 · 2 = 12,56 cm^2.
Halqa: 113,04 − 12,56 = **100,48 cm^2**.

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17. 1,5 · 1,5 = 2,25.
Yuzi = 3,14 · 2,25 = **7,065 m^2**.

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18. 44 cm — diametr. Radius: 44 : 2 = 22 cm.
Yuzi = 3,14 · 22 · 22 = 3,14 · 484 = **1519,76 cm^2**.

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19. Radius: 5,6 : 2 = 2,8 m. Avval 2,8 : 7 = 0,4.
Yuzi = 22 · 0,4 · 2,8 = **24,64 m^2**.

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20. Eni = πr = 3,14 · 4 = **12,56 cm**.
Bo'yi — radius, 4 cm. Yuzi: 12,56 · 4 = 50,24 cm^2.
